

EduVision DV: Dashboard Architecture

Title:

EduVision DV

Higher Education Performance Dashboard

Module 4 — Dashboard Planning & Prototyping

Project Objective: The EduVision dashboard suite provides an interactive platform for analyzing:

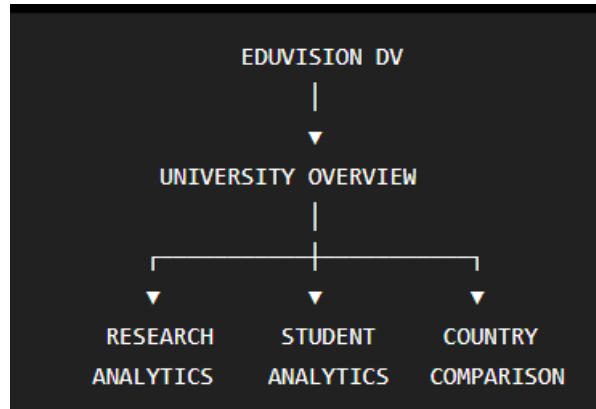
- University rankings
- Academic reputation
- Research performance
- Student characteristics
- Country-level university performance

The final Tableau workbook will contain four interconnected dashboards:

1. University Overview
2. Research Analytics
3. Student Analytics
4. Country Comparison

This matches the project's requirement for a unified, interconnected Tableau dashboard suite.

Dashboard Architecture:



Common Dimensions:

All dashboards will use common dimensions where applicable:

- University
- University ID
- Country
- Country ID
- Region
- Ranking Year
- Source

The project documentation specifically recommends using `university_id` and `country_id` rather than university names alone as relationship keys.

Design Theme

Visual style:

- Dark professional theme
- Purple/blue accent colors
- KPI cards
- Clear section headers
- Consistent navigation
- Minimal visual clutter
- Interactive filters

University Overview:

Dashboard 1: University Overview

Purpose: Provide a high-level view of university performance and allow users to quickly identify top-ranked universities and examine institutional performance.

Wireframe:

EDUVISION DV		Year	Source	Country
NAVIGATION	Global Rank	Ranking	Research	Academic
		Score	Impact	Reputation
	Top Universities by Global Rank			
	Ranking / Score		Universities by	
	Comparison		Region / Country	
Overview	Academic Reputation		Research	
Research	Comparison		Productivity	
Students				
Countries				

KPI Cards:

Display:

- Global Ranking Score
- Research Impact Score
- Academic Reputation Score
- Faculty-to-Student Ratio
- International Student Percentage
- Research Productivity Index

These are the six KPIs required by Module 3.

Main Visualizations:

1. Top Universities by Global Rank

- University
- Country
- Global Rank
- Global Ranking Score

2. University Performance Comparison: Compare selected universities using ranking/KPI measures.

3. Academic Reputation Analysis: Show academic reputation across selected universities.

4. Research Performance: Show Research Impact and Research Productivity.

Filters

- Ranking Year
- Source
- Country
- Region
- University

Dashboard Actions:

University Selection → Filter

Selecting a university updates the related charts.

Navigation → Research Analytics

Selected university should be passed to the Research dashboard.

Navigation → Student Analytics

Selected university should be passed to the Student dashboard.

Navigation → Country Comparison

Selected university's country can be used as the starting country selection.

Research Analytics

Dashboard 2: Research Analytics

Purpose: Analyze research performance, citation-related performance, research impact, and research productivity.

Wireframe:

RESEARCH ANALYTICS		Year	Source	Country
NAVIGATION	Research	Research	Research	
	Impact	Productivity	Score	
	Overview	Research Performance by University		
	Research			
	Students			
	Countries	Research Score		Citation Score
		Research Impact vs Research Productivity		

KPI Cards:

- Research Impact Score
- Research Productivity Index
- Research Score
- Citation Score

Main Visualizations:

1. Research Performance by University: Bar chart showing selected universities and research performance.

2. Citation Performance: Compare citation-related scores across universities.

3. Research Productivity Comparison: Compare Research Productivity Index.

4. Research Impact vs Research Productivity

Scatter plot:

Y-axis → Research Impact Score

X-axis → Research Productivity Index

This allows users to identify universities with strong research impact and productivity.

Filters:

- Year
- Source
- Country
- Region
- University

Interactive Actions:

- Select university → highlight/filter research metrics
- Select country → show universities from that country
- Click Overview → return to University Overview
- Click Students → move to Student Analytics
- Click Countries → move to Country Comparison

Student Analytics

Dashboard 3: Student Analytics

Purpose: Analyze university student population, international student representation, and student/staff indicators.

Wireframe:

STUDENT ANALYTICS				Year	Source	Country
NAVIGATION	Total Students	International Students %	Faculty/Student Ratio			
	Student Population Comparison					
	International Student Percentage		Faculty-to-Student Ratio			
	Student Distribution / Comparison					

KPI Cards:

- Number of Students
- International Student Percentage
- Faculty-to-Student Ratio

Main Visualizations:

- 1. Student Population Comparison:** Compare number of students across universities.
- 2. International Student Percentage:** Compare international student percentage.
- 3. Faculty-to-Student Ratio:** Compare the available student/staff ratio.
- 4. Student Distribution:** Show student-related indicators by country/region where the data supports it.

Filters:

- Year
- Source
- Country
- Region
- University

Interactive Actions:

- University selection → update student metrics
- Country selection → filter universities
- Research button → Research Analytics
- Country button → Country Comparison
- Overview button → University Overview

Country Comparison:

Dashboard 4: Country Comparison

Purpose: Compare university performance across countries and identify differences in ranking, research, and student indicators.

Wireframe:

COUNTRY COMPARISON		Year	Region	Source
NAVIGATION	Universities	Avg Ranking	Avg Research	
		Score	Performance	
	Country Performance Comparison			
	Ranking Comparison		Research Comparison	
	Student / International Comparison			

KPI Cards:

- Number of Universities
- Average Global Ranking Score
- Average Research Impact Score
- Average Research Productivity Index
- Average International Student Percentage

Main Visualizations:

- 1. Country Ranking Comparison:** Compare countries based on university ranking performance.
- 2. Research Performance by Country:** Compare research indicators across countries.
- 3. Student Comparison:** Compare available student indicators.
- 4. Regional Performance:** Compare countries within different regions.

Filters:

- Year
- Region
- Country
- Source

Interactive Actions:

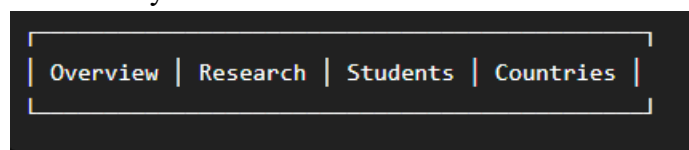
Country Selection: Selecting a country filters the university-level information.

University → Country: From University Overview, selecting a university can transfer its country to this dashboard.

Country Comparison: Allow users to compare multiple countries.

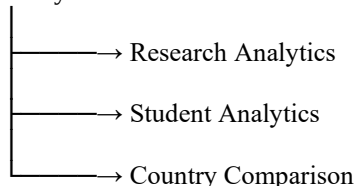
Filters, Navigation & Interactive Actions

Global Navigation: Every dashboard will contain:



Navigation Flow:

University Overview



Users can return to Overview from every dashboard.

Global Filters:

Year: Allows users to select the ranking year.

Source: Allows comparison between:

- QS
- THE
- WUR

Country: Filters universities by country.

Region: Filters universities by geographical region.

University: Allows users to focus on an individual institution.

Dashboard Actions:

Action	Result
Select University	Filters related university information
Select Country	Filters universities within selected country
Select Region	Filters countries/universities in region
Select Year	Updates ranking/performance information
Select Source	Filters ranking-source data
Click Research	Opens Research Analytics
Click Students	Opens Student Analytics
Click Countries	Opens Country Comparison
Click Overview	Returns to University Overview

Interactive User Journey & Validation:

Example User Journey:

1. Open University Overview
- ↓
2. Select University
- ↓
3. View ranking + KPI information
- ↓
4. Click Research Analytics
- ↓
5. Analyze research impact/productivity
- ↓
6. Click Student Analytics
- ↓
7. Analyze student indicators
- ↓
8. Click Country Comparison



9. Compare the university's country with other countries

This is consistent with the project's example of an interlinked user journey from University Overview → Research Analytics → Student Analytics → Country Comparison.